

VIX Regime Classification

Research Summary

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Overview

This project applies the selective predicting methodology from Liu & Jiang (2020) to VIX volatility regime classification. The goal is to predict whether the VIX index will be in a high-volatility regime ($VIX \geq 20$) N trading days ahead, and to show that confidence-based selective predicting consistently improves accuracy over baseline models.

Data & Features

Source	Series	Features	Dataset
Yahoo Finance + FRED	VIX, VIX3M, SP500, Gold, TNY, DXY, SKEW	36 engineered	~5,000 days (1990-present)

Features include VIX technicals (moving averages, Bollinger bands, RSI, momentum), cross-asset signals, S&P 500 equity indicators, and macro factors (Fed Funds Rate, yield curve). Chronological 80/20 train/test split with no shuffling.

Models & Methodology

Random Forest and **XGBoost** (HistGradientBoostingClassifier) with hyperparameter tuning via RandomizedSearchCV ($n_iter=50$, TimeSeriesSplit). Both use isotonic calibration (CalibratedClassifierCV, 5-fold TimeSeriesSplit) for well-calibrated probability estimates.

Selective Predicting: A prediction is made only when model confidence exceeds threshold t (i.e. $P > 0.5 + t$ or $P < 0.5 - t$). Uncertain days near 0.5 are abstained from, trading coverage for higher accuracy.

Main Results ($VIX \geq 20$, $t = 0.25$)

N (days ahead)	RF Baseline	RF Selective	Gain	Coverage
5	83.8%	91.1%	+7.3%	78.4%
10	78.6%	88.8%	+10.2%	70.5%
15	74.9%	90.0%	+15.1%	58.1%
20	72.6%	90.9%	+18.3%	50.6%
25	69.7%	91.1%	+21.5%	32.9%

N (days ahead)	XGB Baseline	XGB Selective	Gain	Coverage
5	82.9%	90.7%	+7.8%	79.4%
10	79.0%	91.0%	+12.0%	61.1%

15	73.5%	87.7%	+14.2%	54.6%
20	74.4%	92.1%	+17.7%	35.4%
25	66.5%	92.5%	+26.0%	24.1%

Confidence-based selective predicting yields consistent accuracy gains across all prediction horizons. Gains increase with N - at 25 days ahead, XGBoost improves from 66.5% to 92.5%.

Robustness - Multiple VIX Thresholds (RF Accuracy Gain)

N	VIX >= 18	VIX >= 20	VIX >= 22
5	+7.6%	+9.0%	+8.9%
10	+11.1%	+8.9%	+10.7%
15	+16.6%	+10.8%	+11.4%
20	+15.2%	+15.1%	+13.1%
25	+25.9%	+8.9%	+9.1%

All gains are positive across every VIX threshold. Result is robust.

Sustained Regime Labels

A harder variant: predict whether VIX >= 20 for **3 consecutive days** starting at day N. Models perform *better* on this task - sustained regimes have stronger precursors.

N	RF Sustained Selective	RF Instantaneous Selective
5	94.0%	91.1%
10	93.8%	88.8%
15	93.8%	90.0%
20	94.3%	90.9%
25	89.9%	91.1%

Walk-Forward Validation (5 Expanding Windows)

Model	N	Mean Baseline	Mean Selective	Delta
Random Forest	5	82.8% +/- 9.7%	92.2% +/- 1.7%	+9.4%
XGBoost	5	84.0% +/- 8.0%	84.6% +/- 9.7%	+0.6%
Random Forest	10	72.3% +/- 22.7%	85.9% +/- 7.7%	+13.7%

RF selective predicting is notably more consistent across time (lower std dev) than the baseline.

Feature Importance (Random Forest, avg across N = 5, 10, 20)

Rank	Feature	Avg Importance	Category
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1	VIX_SMA10	19.8%	VIX trend
2	VIX_SMA20	15.3%	VIX trend
3	VIX_BB_UPPER	10.9%	VIX Bollinger bands
4	VIX_BB_LOWER	10.8%	VIX Bollinger bands
5	SP500_DRAWDOWN	6.8%	Equity risk
6	SP500_RVOL	5.7%	Realized volatility
7	FEDFUNDS_CHANGE3M	2.2%	Macro
8	VIX_BB_STD	2.2%	VIX vol-of-vol

VIX trend/level features account for ~57% of predictive power. Macro features grow more important at longer prediction horizons.

GitHub Repository

https://github.com/akshatg10104/VIX_research